

National Technical University of Athens
School of Electrical and Computer Engineering (ECE)

10th semester

Control Systems of Electrical Machines

Simulation Assignment 1: Modeling and control of a DC machine

Dimitris Vatalas
(Student ID: 03121093)

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1 DC Machine Model

The dynamics of the direct current (DC) machine are described by the following system of state equations:

$$L\frac{di}{dt} = v - Ri - \psi\omega_r \tag{1}$$

$$J\frac{d\omega_r}{dt} = \psi i - \tau_L \tag{2}$$

The machine parameters used are presented in the table (1) below:

Quantity	**Value**
Rated voltage V_N	120 V
Rated current I_N	20 A
Rated speed n_N	3000 rpm
Rated torque τ_N	7 Nm
Armature resistance R	0.5 Ω
Armature inductance L	2.5 mH
Flux linkage ψ	0.35 Vs
Moment of inertia J	0.001 kgm^2

Table 1: Parameters of the DC machine

The model used in Simulink is shown in Figures (1) and (2).

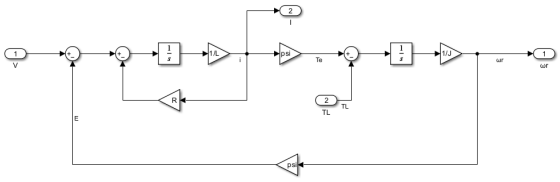


Figure 1: DC Motor model

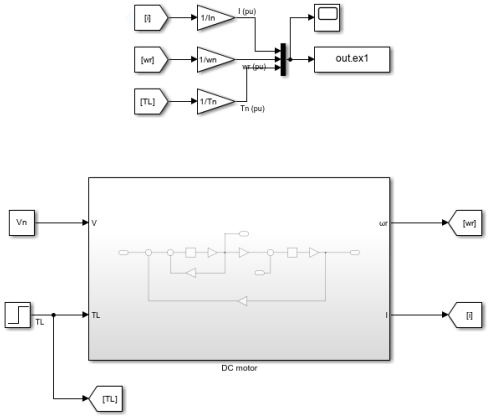


Figure 2: System schematic

Simulation Results

Figures (3) and (4) present the current and speed responses respectively, along with the variation of the load torque. It is noted that these waveforms, as well as all other figures in the project, are expressed in normalized (per-unit) form based on the parameters of Table (1).

The system is supplied with rated voltage $v = V_N$. The armature current increases abruptly and then decreases, reaching zero since the load torque is zero. The speed ω_r increases gradually and reaches steady state at a value higher than the rated one, approximately 1.1 *p.u.*.

When the system reaches steady state, at 100 ms, a load torque $\tau_L = \tau_N$ is applied, causing an increase in current and a drop in rotational speed. It is observed that both quantities, after the transient response, settle at their rated values, 1 $p.u.$, as expected.

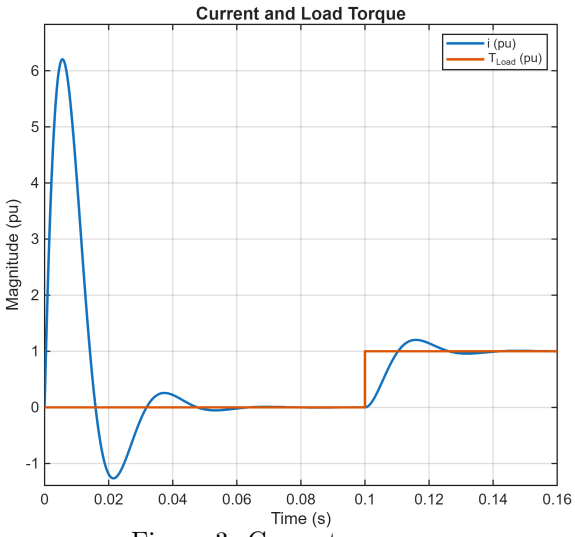


Figure 3: Current response

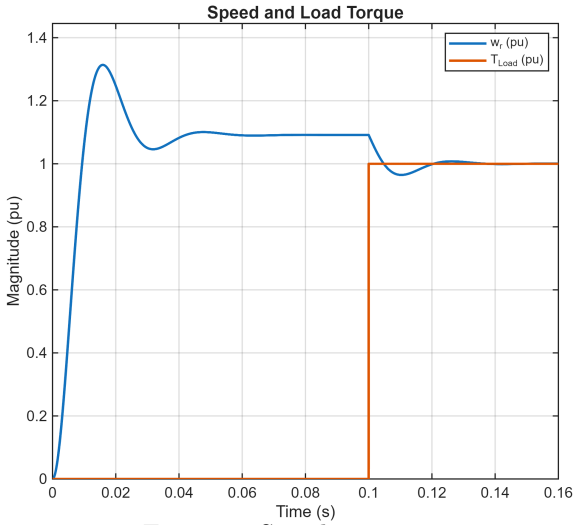


Figure 4: Speed response

Explanation of Current Peak

The large current peak at startup is due to the fact that initially the rotor speed is zero ($\omega_r = 0$). This means that the back-emf ($\psi\omega_r$) is zero and the state equation becomes:

$$L\frac{di}{dt} = v - Ri \tag{3}$$

This leads to a very large initial current value, exceeding 6 $p.u.$. As the speed increases, the term $\psi\omega_r$ in equation (1) increases, thus reducing the overall voltage across the windings and consequently decreasing the current in steady state.

2 Current Control

In this section, current control of the machine is implemented using a two-degrees-of-freedom PI controller, along with an anti-windup mechanism to prevent integral windup.

The controller is described by the following equations:

$$e = i_{\text{ref}} - i \tag{4}$$

$$v_{\text{ref}} = k_p e + k_i I - R_a i + \hat{E} \tag{5}$$

$$\bar{v}_{\text{ref}} = \text{sat}(v_{\text{ref}}, V_{\text{base}}) \tag{6}$$

$$I = \frac{1}{s}\left(e + \frac{1}{k_p}(\bar{v}_{\text{ref}} - v_{\text{ref}})\right) \tag{7}$$

Next, the physical meaning of each controller equation is presented:

- Current error (4): The quantity e represents the error between the desired current i_{ref} and the actual current i. The objective of the controller is to drive this error to zero.

- PI equation with feedforward (5): This equation represents the classical PI controller. The controller has two degrees of freedom, as it includes additional terms that do not depend solely on the error:
  - The term $R_a i$ introduces the active resistance, which improves the dynamic behavior of the system.
  - The term $\hat{E} = \psi\,\omega_r$ is the feedforward of the back-EMF. If the speed ω_r is known or estimated, this term compensates for the disturbance caused by the back-EMF, improving the system response.
- Saturation (6): The controller output is limited through a saturation function to account for the voltage limits of the converter and the machine.
- Integrator with anti-windup (7): This term corresponds to the integral action of the PI controller with an anti-windup mechanism using the back-calculation method.

  Specifically, when the controller enters saturation ($v_{\text{ref}} \neq \bar{v}_{\text{ref}}$), the integrator is corrected so that it does not continue to increase uncontrollably. In this way, integral windup is avoided, which would otherwise lead to large overshoot and poor transient response.

The rise time $t_{\text{rise}} = 1$ ms of the closed-loop system and the system bandwidth α_c are given by:

$$\alpha_c = \frac{\ln 9}{t_{\text{rise}}} = \frac{\ln 9}{0.001} = 2197.22 \tag{8}$$

Thus, the controller gains are:

$$k_p = a_c L = 5.4931 \tag{9}$$

$$k_i = a_c^2 L = 12069 \tag{10}$$

$$R_a = a_c L - R = 4.9931 \tag{11}$$

The current controller model was implemented in Simulink and is shown as a subsystem in Figure (5), while the overall simulation schematic is shown in Figure (6).

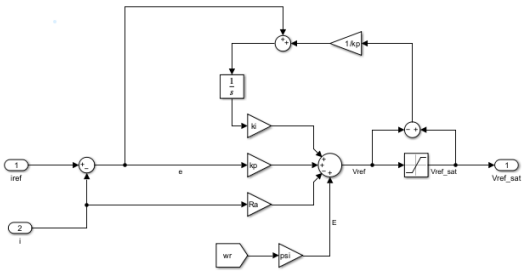


Figure 5: Current controller

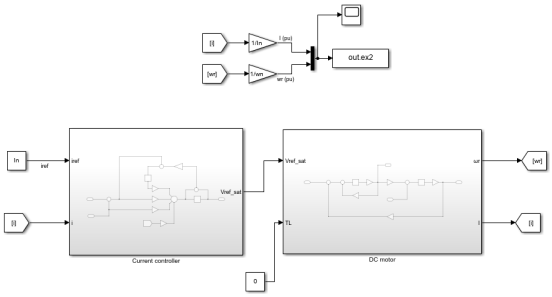


Figure 6: System schematic

Simulation Results

Figure (7) shows the current and speed response under no-load conditions when the reference current is equal to the rated value. It is observed that the current remains within the limits of its rated value and that the speed stabilizes at a value slightly higher than the rated one (1 *p.u.*).

Figure (8) shows the current variation at the beginning of the simulation in order to verify the rise time. Specifically, the rise time t_r is defined as the time required for the system response to increase from 10% to 90% of its final value. In this case, it is observed that:

$$t_r = t_{90\%} - t_{10\%} \approx 1.0724 - 0.0483 = 1.024\ msec \tag{12}$$

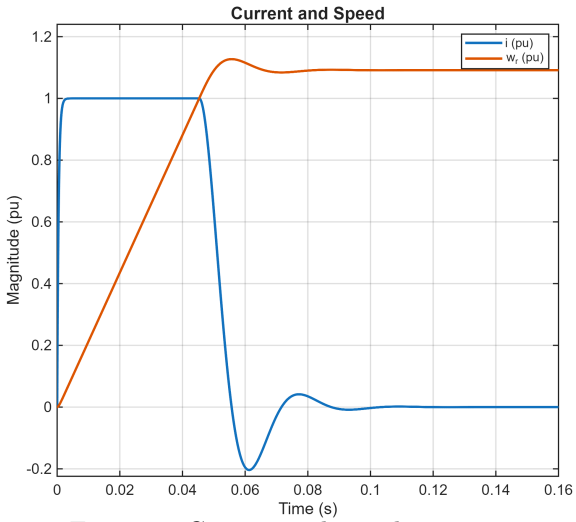


Figure 7: Current and speed response

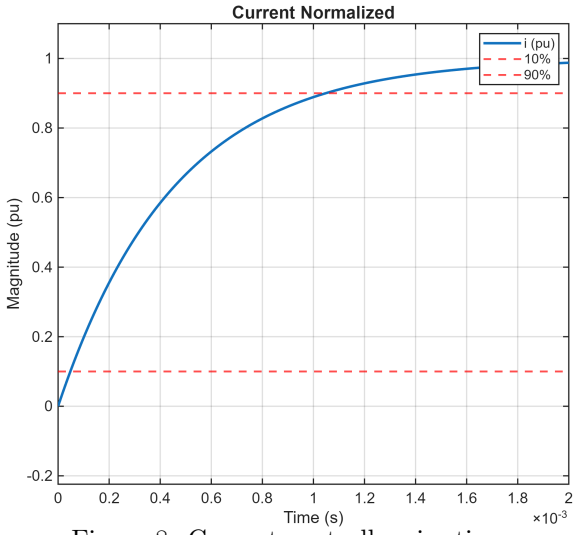


Figure 8: Current controller rise time

3 Speed Control

In the next step, speed control of the machine is implemented using a two-degrees-of-freedom PI controller. The controller gains are selected such that the speed rise time is ten times greater than the corresponding rise time of the current controller, i.e.:

$$t_{\text{rise,s}} = 10\, t_{\text{rise,c}} = 10 \text{ ms} \tag{13}$$

The speed controller is described by the following equations:

$$e_s = \omega_{\text{ref}} - \omega_r \tag{14}$$
$$i_{\text{ref}}^{\text{nom}} = k_{ps} e_s + k_{is} I_s - b_a \omega_r \tag{15}$$
$$i_{\text{ref}} = \text{sat}(i_{\text{ref}}^{\text{nom}}, I_{\max}) \tag{16}$$
$$I_s = \frac{1}{s}\left(e_s + \frac{1}{k_{ps}}(i_{\text{ref}} - i_{\text{ref}}^{\text{nom}})\right) \tag{17}$$

The physical meaning of the terms is similar to that of the current controller and is presented below:

- Speed error (14): The quantity e_s is the difference between the desired speed ω_{ref} and the actual speed ω_r.
- PI equation (15): The controller is a classical PI controller with two degrees of freedom, including the term $b_a\omega_r$, which does not depend on the error e_s, in order to improve the response to load disturbances.
- Saturation (16): The controller output is limited according to the current limits of the DC machine or the converter.
- Integrator with anti-windup (17): This term corresponds to the integral action of the PI controller with an anti-windup mechanism using the back-calculation method.

The desired rise time and the corresponding bandwidth of the speed controller are calculated from the relation:

$$\alpha_s = \frac{\ln 9}{t_{\text{rise},s}} = \frac{\ln 9}{0.01} = 219.72 \tag{18}$$

Thus, the speed controller gains are:

$$k_{ps} = \alpha_s \frac{J}{\psi} = 0.6278 \tag{19}$$

$$k_{is} = \alpha_s^2 \frac{J}{\psi} = 137.937 \tag{20}$$

$$b_{as} = \frac{\alpha_s J - b}{\psi} = \frac{\alpha_s J}{\psi} = 0.6278 \tag{21}$$

The speed controller model was implemented in Simulink and is shown in Figure (9). The overall simulation schematic is shown in Figure (6) with the new controller integrated.

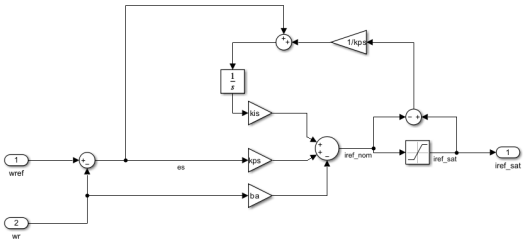


Figure 9: Speed controller sub-system

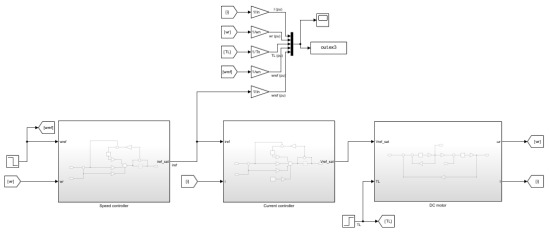


Figure 10: System schematic with speed controller

Simulation Results

To validate the performance, a step change in the speed reference of 160 rad/s = 0.5093 $p.u.$ was applied at 0.07 sec, along with a step increase in load from zero to nominal at 0.15 sec.

Figures (11) and (12) show the current and speed ω_r responses respectively, with respect to the reference ω_{ref} and the load $\tau_L = \tau_N$.

It is observed that both controllers (current and speed) track their references with high accuracy, keeping the current and rotational speed within the machine limits. Specifically, regarding the current, it is observed that when the speed reference changes, the speed controller provides a negative current reference (-1 $p.u.$) to the current controller in order to brake the motor within the specified rise time limits. As the motor approaches the desired speed, the reference decreases in magnitude, reaching zero when the motor reaches the reference speed. During the step change in load, the current increases to its nominal value.

Regarding the speed response during the described transients, it is observed that the controller tracks the speed reference without oscillations. It is worth noting that when nominal load is applied, the motor is unable to reach the reference speed ($\omega_{ref} = 1 - 0.5093 = 0.4907$ $p.u.$). This occurs because the transient event, i.e., the step increase in load, is faster than the system rise time. Thus, as soon as the transient occurs, the current increases to its nominal value and stabilizes there, providing sufficient torque to counteract the load. Since a saturation block is included, the current cannot exceed its nominal value in order for the motor to provide additional torque and reach the nominal speed.

In Figures (13) and (14), the simulation results are shown with an increased maximum allowable current in the saturation block, specifically increased to 1.5 $p.u.$. In this case, it is observed that under maximum load conditions, the controller accelerates and manages to reach the reference speed.

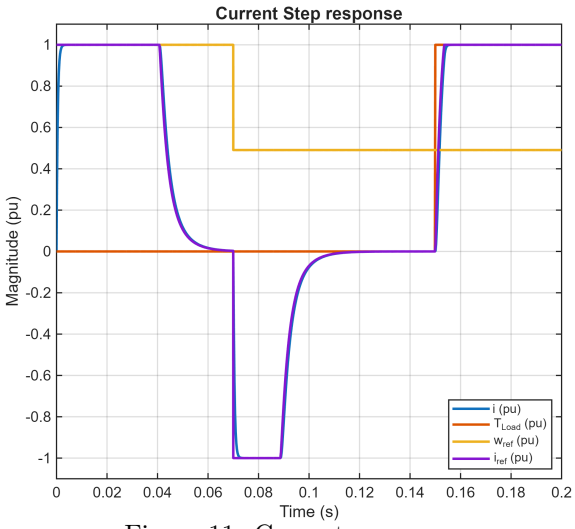


Figure 11: Current response

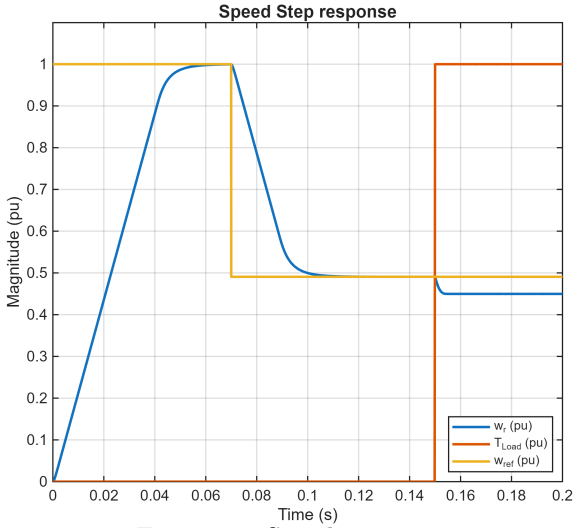


Figure 12: Speed response

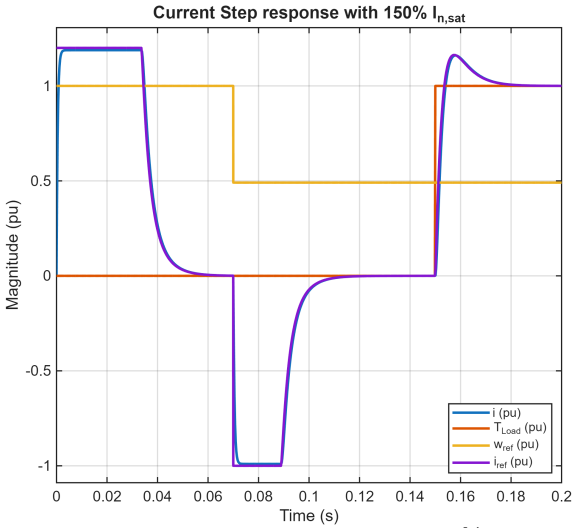


Figure 13: Current response with 150% $I_{n,sat}$

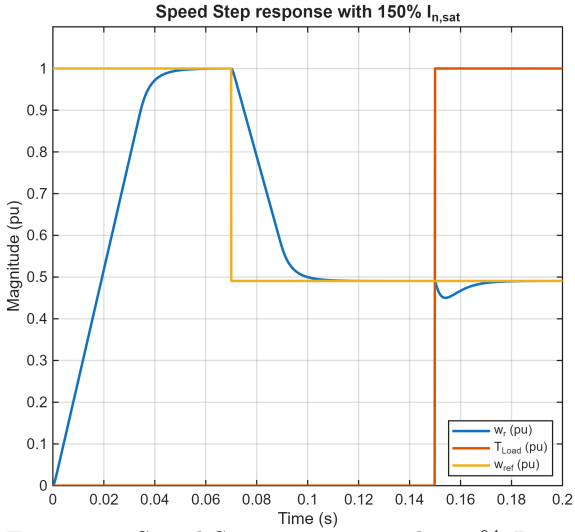


Figure 14: Speed Step response with 150% $I_{n,sat}$

Next, in Figure (15), the variation of the rotational speed at the beginning of the simulation is shown in order to verify the rise time. Specifically, the rise time $t_{r,s}$ is:

$$t_{r,s} = t_{90\%} - t_{10\%} \approx 40.548 - 4.954 = 35.594 \; msec \tag{22}$$

It is observed that the rise time differs significantly from the theoretically calculated value $t_{r,s} = 10 \; msec$. This is due to current and voltage saturation through the saturation block, i.e., the outputs of both PI controllers are limited, resulting in lower current, lower torque, lower acceleration, and therefore a larger speed rise time. In Figure (16), the rise time of the system is shown when these saturations are removed. In this case, the rise time is very close to its theoretical value:

$$t_{r,s} = t_{90\%} - t_{10\%} \approx 9.9726 - 0.8253 = 9.1473 \; msec \tag{23}$$

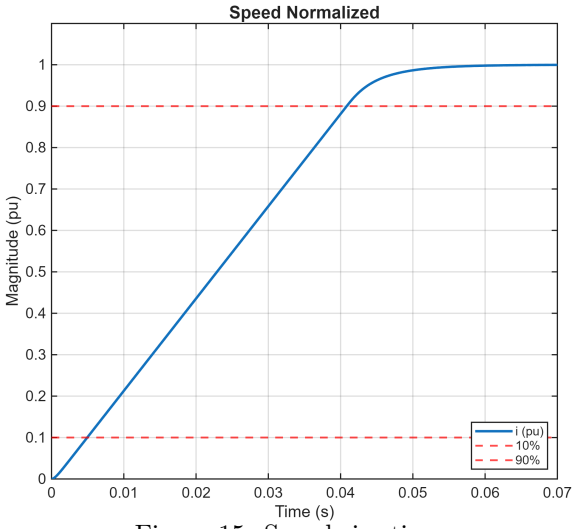


Figure 15: Speed rise time

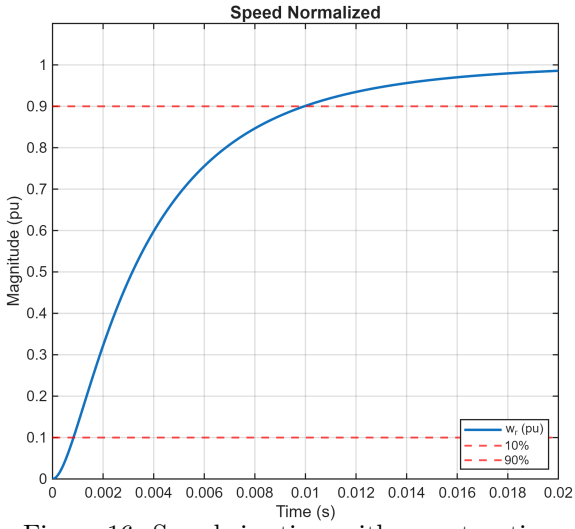


Figure 16: Speed rise time with no saturation

4 Single Degree of Freedom Speed Control

In the next step, the speed controller is modified to a single degree of freedom controller. The main difference compared to the two-degrees-of-freedom controller is that the term b_a is removed and the control action is based exclusively on the error. Therefore, equation (15) becomes:

$$i_{\text{ref}} = k'_{ps} e_s + k'_{is} I_s \tag{24}$$

The new controller gains (k'_{ps}, k'_{is}) are calculated so as to maintain the same response time ($t_{rise,s} = 10\ msec$). The system dynamics are given by:

$$\frac{d\omega_r}{dt} = \frac{\psi}{J} i - \frac{\tau_L}{J} \tag{25}$$

Applying the Laplace transform to the full system equation yields:

$$s\Omega_r(s) = \frac{\psi}{J} I(s) - \frac{1}{J} T_L(s) \tag{26}$$

For the transfer function, the load disturbance is assumed to be zero ($T_L(s) = 0$), thus:

$$s\Omega_r(s) = \frac{\psi}{J} I(s) \tag{27}$$

Therefore, the transfer function is:

$$G_s(s) = \frac{\Omega_r(s)}{I(s)} = \frac{\psi}{J} \cdot \frac{1}{s} \tag{28}$$

A first-order system is selected as the desired closed-loop transfer function:

$$G_{sc}(s) = \frac{a_s}{s + a_s} \tag{29}$$

According to the direct synthesis method, the controller is given by:

$$F_s(s) = \frac{G_{sc}(s)}{G_s(s)\left(1 - G_{sc}(s)\right)} \tag{30}$$

Thus:

$$F_s(s) = \frac{a_s}{s} G_s^{-1}(s) = \frac{a_s}{s} \frac{sJ}{\psi} = \frac{Ja_s}{\psi} \quad \text{and} \quad F_s(s) = k'_p \tag{31}$$

Therefore, the resulting controller is purely proportional (with zero integral action):

$$k'_{ps} = \frac{Ja_s}{\psi} = 0.6278 \tag{32}$$

$$k'_{is} = 0 \tag{33}$$

Simulation Results

To evaluate the controller performance, the same experiment as before is repeated, i.e., a step change in the speed reference of 160 $rad/s = 0.5093$ $p.u.$ is applied at 0.07 sec, along with a step increase in load from zero to nominal at 0.15 sec.

Figures (17) and (18) show the current and speed ω_r responses respectively, with respect to the reference ω_{ref} and the load $\tau_L = \tau_N$.

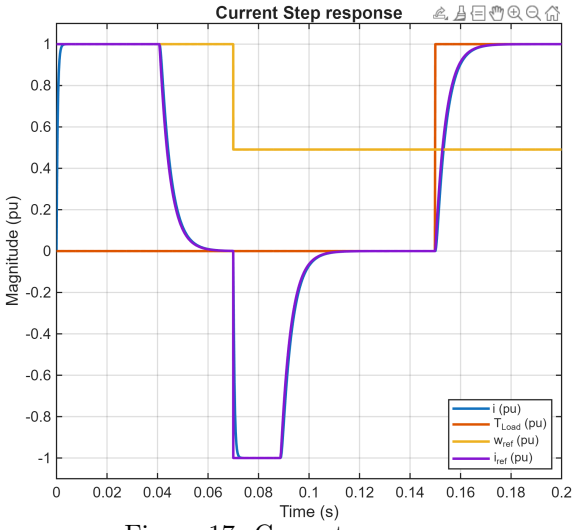


Figure 17: Current response

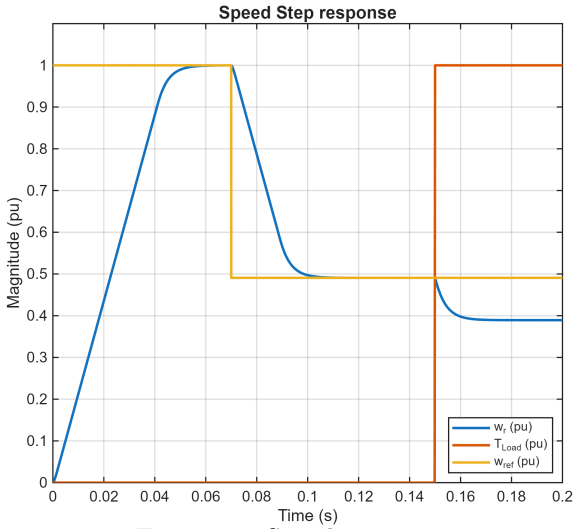


Figure 18: Speed response

It is observed that both controllers again track their references during the transient phenomena. The only notable difference appears in the steady-state speed after the load transient. Specifically, in this case the machine speed stabilizes around 0.4 $p.u.$, whereas in the two-degrees-of-freedom case it reached approximately 0.46 $p.u.$.

To further validate the controller operation, the experiment is repeated with an increased maximum allowable current in the saturation block, specifically with the saturation current set to 1.5 $p.u.$. The simulation results are shown in Figures (19) and (20).

It is observed that in this case, under maximum load conditions, the controller is unable to reach the reference speed, unlike the corresponding two-degrees-of-freedom case shown in Figure (14). This occurs because the single-degree-of-freedom controller does not include a separate feedforward term to compensate for the load disturbance. As a result, even though the machine is capable of producing additional torque, the speed error is not eliminated.

Figures (21) and (21) show the variation of the rotational speed at the beginning of the simulation in order to verify the rise time with and without saturation, respectively. Specifically, it is observed that the rise time remains approximately the same as in the previous case, while the rise time without current and voltage saturation is very close to the theoretical value, as expected.

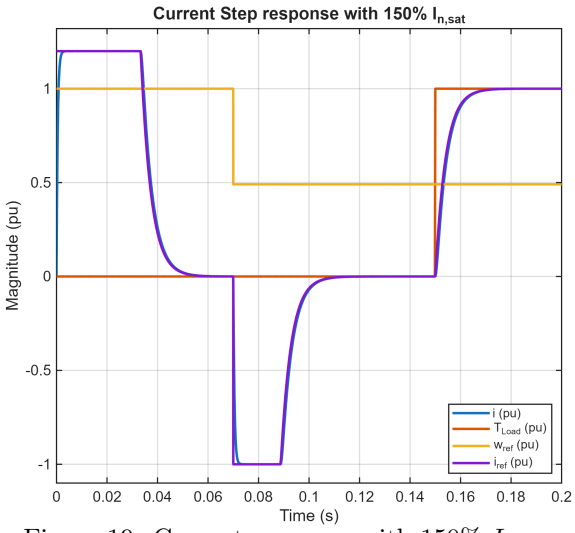


Figure 19: Current response with 150% $I_{n,sat}$

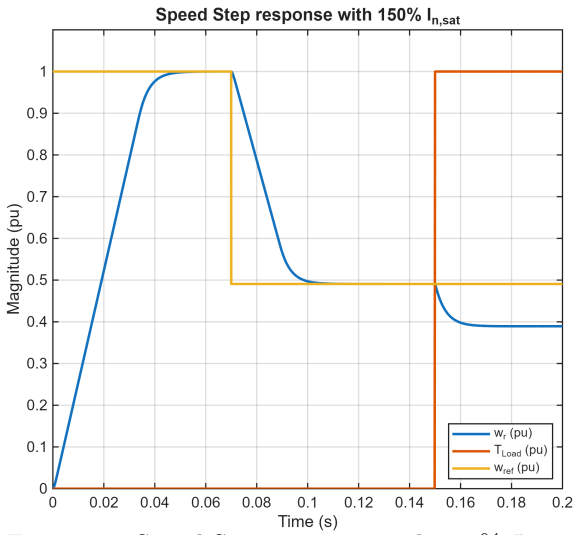


Figure 20: Speed Step response with 150% $I_{n,sat}$

With current and voltage saturation:

$$t_{r,s} = t_{90\%} - t_{10\%} \approx 40.548 - 4.954 = 35.594\ msec \tag{34}$$

Without current and voltage saturation:

$$t_{r,s} = t_{90\%} - t_{10\%} \approx 9.8814 - 0.8199 = 9.62\ msec \tag{35}$$

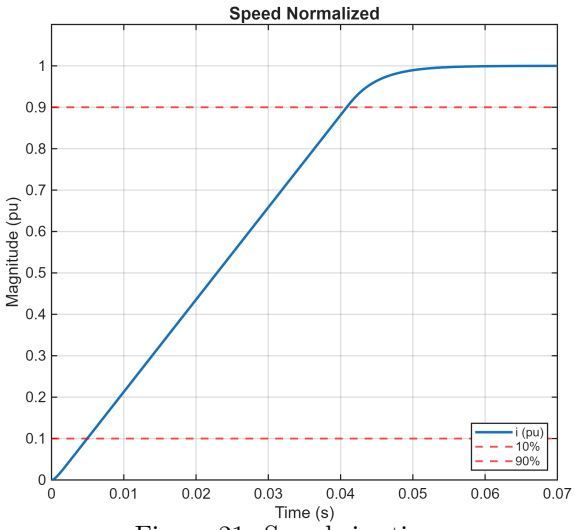


Figure 21: Speed rise time

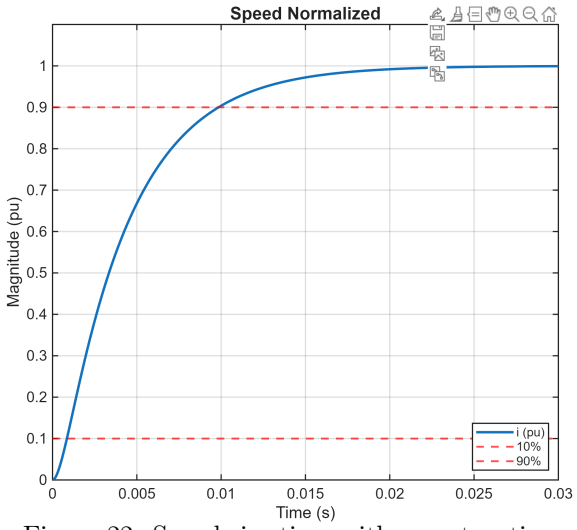


Figure 22: Speed rise time with no saturation

5 Deactivation of anti-windup

In this task, the effect of disabling the anti-windup mechanism in the speed and current controllers is examined, without removing the saturation of the controller output.

Speed controller

The main modification in the speed controller is the removal of the back-calculation term, i.e., the term $\frac{1}{k_{ps}}(i_{\text{ref}} - i_{\text{ref}}^{\text{nom}})$. Therefore, all equations stay the same apart from equation (17), which becomes:

$$I_s = \frac{1}{s} e_s \tag{36}$$

Simulation results

In this experiment, the speed reference is initially set to its nominal value, and at 0.35 *secs* it is changed to half of the nominal rotational speed.

Figure (23) presents the speed response with and without anti-windup. It is observed that without anti-windup, the system exhibits a significantly large overshoot, reaching 1.1 *p.u.*, and also requires considerably more time to settle at the reference value. This occurs due to the accumulation of integral error, as shown in Figure (24).

Specifically, while the motor is stationary, i.e., at the beginning of the simulation, the error at the output of the integrator continuously increases as it keeps accumulating. When the motor exceeds the reference speed, the error begins to decrease at a slow rate until it reaches zero. However, when anti-windup is used, the integrator is dynamically corrected whenever the controller output enters saturation, preventing the continuous accumulation of error that causes the overshoot. As a result, the speed reaches its reference faster and more smoothly, without significant overshoot.

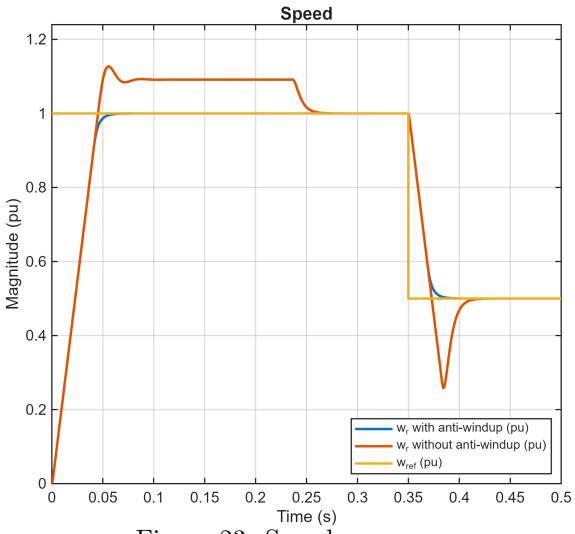


Figure 23: Speed response

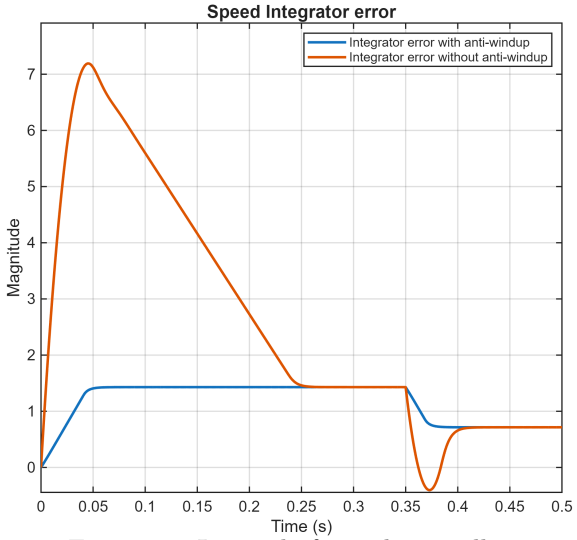


Figure 24: Integral of speed controller

Current controller

In this experiment, the anti-windup mechanism is restored in the speed controller and removed from the current controller, i.e., the back-calculation term $\frac{1}{k_p}(\bar{v}_{\text{ref}} - v_{\text{ref}})$ is eliminated. Therefore, all equations stay the same apart from equation (7), which becomes:

$$I = \frac{1}{s} e \tag{37}$$

Simulation results

The speed reference is set to its nominal value as before, and a transient is introduced at 100 $msec$ by applying nominal mechanical load. The system response is then observed with and without anti-windup.

Figure (25) shows the current response with and without the anti-windup scheme, along with the current reference, i.e., the output of the speed controller. It is observed that without anti-windup, the current exhibits a significant overshoot at the moment of the transient, exceeding 1.3 $p.u.$. As before, this overshoot is due to the accumulation of integral error, as illustrated in Figure (26).

Specifically, upon load application, the current error continuously accumulates in the integrator, leading to an excessively high output. With anti-windup enabled, the difference between the saturated and actual output is continuously subtracted from the integrator, limiting the overshoot and allowing the current to quickly settle at the reference value.

It is worth noting that in this case the error is eliminated much faster compared to the speed controller case, which is expected due to the faster response of the current controller.

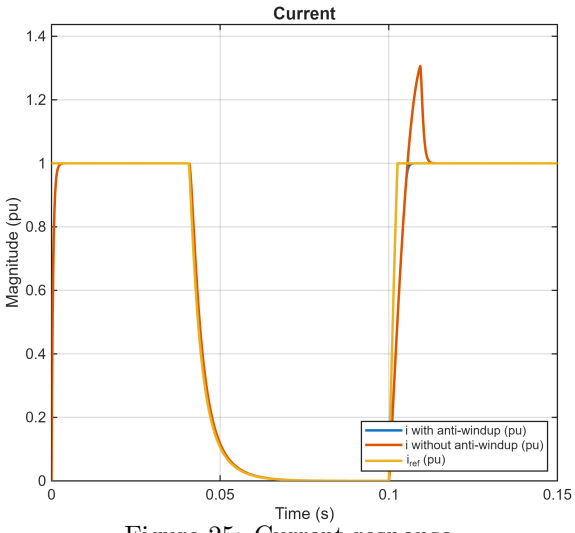


Figure 25: Current response

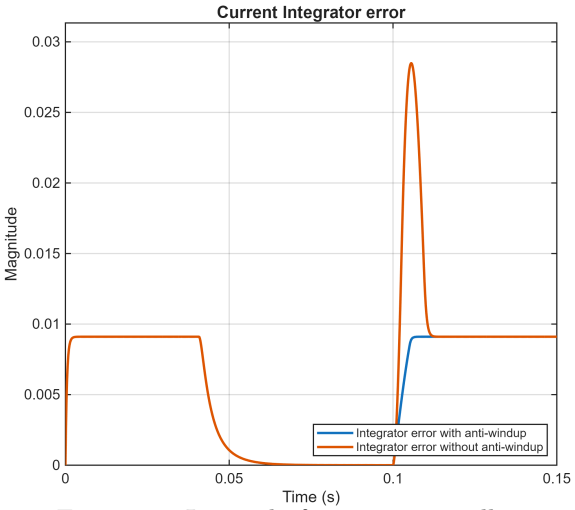


Figure 26: Integral of current controller